



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

April 2014

Assessment Task 2
Year 12

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 80

- Attempt sections A – D.
- Start each **NEW** section in a separate answer booklet.
- Hand in your answers in 4 separate bundles:
Section A
Section B
Section C
Section D

Examiner: *J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

START A NEW ANSWER BOOKLET

SECTION A [20 marks]

Marks

1. If $f(x) = x + 5$, find the inverse function $f^{-1}(x)$.

[1]

2. Find the Cartesian equation whose parametric equations are

[1]

$$x = 8t, \quad y = 4t^2$$

3. If α, β, γ are the roots of the equation $x^3 + x^2 - 2x - 1 = 0$. Find the value of

(i) $\alpha + \beta + \gamma$

[1]

(ii) $\alpha\beta\gamma$

[1]

(iii) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

[1]

4. Let $y = e^x$. Find $\frac{dy}{dx}$ when $x = 1$.

[2]

5. Find

[2]

$$\frac{d}{dx} \cos^{-1} 2x$$

6. There are 40 Students in a class. 38 of them are boys. If 5 students are randomly selected, find the number of ways that at least 1 girl is selected.

[2]

7. Let $P(x)$ be a polynomial. When $P(x)$ is divided by $x - 2$, the quotient is $6x^2 + 5x + 9$. It is given that $P(2) = 20$.

(i) Find $P\left(\frac{2}{3}\right)$.

[2]

(ii) Solve $P(x) = 0$

[3]

8. Peter invites 8 friends to join his birthday party. There are 5 boys and 4 girls at the party including Peter. In a game, boys and girls sit in a row. What is the probability that the boys and girls sit alternatively?

[2]

9. Find

[2]

$$\int \frac{2}{\sqrt{16 - x^2}} \cdot dx$$

End of Section A

START A NEW ANSWER BOOKLET

SECTION B [20 marks]

Marks

1. Let $P(x) = x^3 + x^2 + mx + n$. If $P(1) = 0$ and the remainder of $P(x)$ divided by $x - 2$ is 11, then factorise $P(x)$.

[3]

2. Consider the function $f(x) = \sin^{-1}(x - 2)$, evaluate $f\left(\frac{3}{2}\right)$.

[2]

3. There are 24 boys and 4 girls in a class. From the class, 5 students are randomly selected to form the class committee.

- (i) Find the probability that the class committee consists of boys only.

[1]

- (ii) Find the probability that the class committee consists of at least 2 boys and 1 girl.

[3]

4. Show that if $t = \tan x$, then

[2]

$$\frac{dx}{dt} = \frac{1}{1 + t^2}$$

5. Find

- (i)

$$\int \frac{1}{\sqrt{25 - 9x^2}} \cdot dx$$

[2]

- (ii)

$$\int \frac{x + 1}{x^2 + 1} dx$$

[2]

6. Find

[2]

$$\int (e^x + e^{-x})^2 \cdot dx$$

7. Find the volume of the solid generated when the region bounded by the curve $y = e^x$ and the x -axis in the interval $0 \leq x \leq 2$ is rotated about the x -axis.

[3]

End of Section B

START A NEW ANSWER BOOKLET

SECTION C [20 marks]

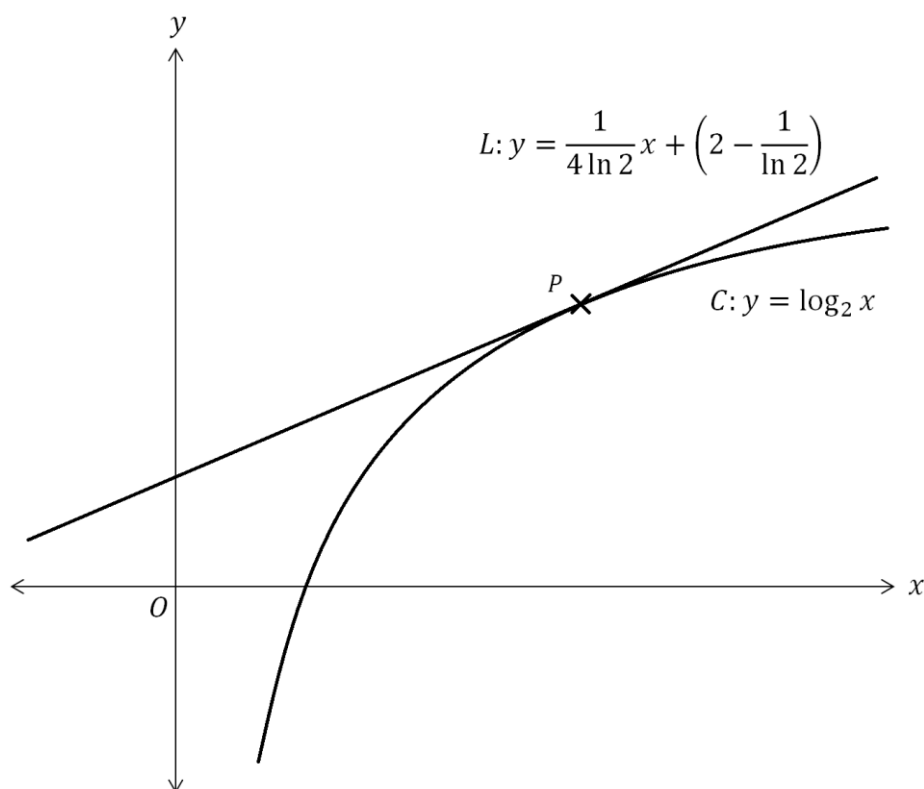
Marks
[2]

1. Find

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{\sqrt{3-4x^2}}$$

2. In the diagram below, the line $L: y = \frac{1}{4 \ln 2} x + \left(2 - \frac{1}{\ln 2}\right)$ touches the curve $C: y = \log_2 x$ at the point P. Find the coordinates of P.

[3]



3. Prove, by mathematical induction, that $9^n - 1$ is divisible by 8 for all positive integers n .
4. The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = 2e^{2x} - 3e^x$. The y -intercept of the curve is 0. Find the area bounded by the curve and the x -axis.

[3]

[4]

Section C continues on next page

5.

(i) Find $\frac{d}{dx}(x^2 \ln x)$

[2]

(ii) Hence, find

[3]

$$\int x \ln x . dx$$

6. Sketch $y = 3 \cos^{-1} 2x$. State the domain and range.

[3]

End of Section C

START A NEW ANSWER BOOKLET

SECTION D [20 marks]

Marks

1. $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.
The tangents at P and Q meet at the point T .

(i) Show that the equation of the tangent at P is $y = px - ap^2$ and similarly, write down the equation of the tangent at Q . [2]

(ii) Find the coordinates of T . [2]

(iii) Let M be the midpoint of PQ , find the coordinates of M . [1]

(iv) The point R is the midpoint of MT , show that R has coordinates [1]

$$\left[a(p+q), \frac{a(p+q)^2}{4} \right]$$

(v) Find the Cartesian equation of the locus of R and describe the locus in geometrical terms. [2]

2. Consider the following letters [3]

A, A, B, B, C, C

how many different arrangements are possible if no two identical letters are next to one another?

3.

(i) Show that [2]

$$a^n + b^n = (a+b)(a^{n-1} + b^{n-1}) - ab(a^{n-2} + b^{n-2}),$$

where a and b are real numbers.

(ii) Let $x + \frac{1}{x} = 2 \cos \theta$, where $\theta \neq \pi$ and $x \neq 0$. Suppose [3]

$$x^k + \frac{1}{x^k} = 2 \cos k\theta \text{ and } x^{k+1} + \frac{1}{x^{k+1}} = 2 \cos(k+1)\theta \text{ for a positive integer } k.$$

$$\text{Show that } x^{k+2} + \frac{1}{x^{k+2}} = 2 \cos(k+2)\theta.$$

(iii) It is given that $x^n + \frac{1}{x^n} = 2 \cos n\theta$. Prove, by mathematical induction, that [4]

$$(x + x^3 + x^5 + \cdots + x^{2n-1}) + \left(\frac{1}{x} + \frac{1}{x^3} + \frac{1}{x^5} + \cdots + \frac{1}{x^{2n-1}} \right) = \frac{\sin 2n\theta}{\sin \theta}$$

for all positive integers n , where $\theta \neq k\pi$.

End of Section D

End of Exam

Section A

1. $f(x) = x + 5$

ie $y = x + 5$

For $f^{-1}(x)$:

$x = y + 5$

ie $y = x - 5$

So $f^{-1}(x) = x - 5$ [1]

2. $x = 8t$; $y = 4t^2$

$t = \frac{x}{8}$; $y = 4\left(\frac{x}{8}\right)^2$

$= \frac{4x^2}{64}$

$\therefore y = \frac{x^2}{16}$ [1]

3. $x^3 + x^2 - 2x - 1 = 0$

(i) $\alpha + \beta + \gamma = -1$ [1]

(ii) $\alpha\beta\gamma = 1$ [1]

(iii) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

$= \alpha\beta\gamma(\alpha + \beta + \gamma)$

$= +1 \times -1$

$= -1$ [1]

4. $y = e^x$

$\frac{dy}{dx} = e^x$

When $x = 1$

$\frac{dy}{dx} = e^1$

$= e$ [2]

5. $\frac{d}{dx} \cos^{-1} 2x$

$= \frac{-1}{\sqrt{1-(2x)^2}} \times 2$

$= \frac{-2}{\sqrt{1-4x^2}}$ [2]

6. No. of ways of choosing 5 including at least one girl is the total no. of ways of choosing 5, minus the no. of ways of choosing 5 boys.

$40C_5 - 38C_5$

$= 156066$

[2]

7. Division transformation: (2)

$$P(x) = (x-2)(6x^2+5x+9) + 20$$

$$(i) P\left(\frac{2}{3}\right) = 0 \quad [2]$$

$$(ii) P(x) = 6x^3 - 7x^2 - x + 2$$

From (i) $\left(x - \frac{2}{3}\right)$ is a factor

$$P(1) = 0$$

$\therefore (x-1)$ is a factor

$$\text{Now } \alpha \beta \gamma = -\frac{2}{6}$$

$$= -\frac{1}{3}$$

$$\therefore \frac{2}{3} \times 1 \times \gamma = -\frac{1}{3}$$

$$\gamma = -\frac{1}{3} \times \frac{3}{2}$$

$$= -\frac{1}{2}$$

$$\therefore P(x) = 0 \text{ for } x = -\frac{1}{2}, \frac{2}{3}, 1 \quad [3]$$

8.

B	B	B	B	B
---	---	---	---	---

$$\text{Probability} = \frac{5! \times 4!}{9!}$$

$$= \frac{1}{126}$$

[2]

$$9. \int \frac{2}{\sqrt{16-x^2}} dx$$

$$= 2 \int \frac{dx}{\sqrt{4^2-x^2}}$$

$$= 2 \sin^{-1} \frac{x}{4} + C \quad [2]$$

(X1)

SECTION B

(1) $P(1) = 0 \Rightarrow 1 + 1 + m + n = 0$
ie $\boxed{m + n = -2} \quad \text{--- (1)}$

Also $P(2) = 11 \Rightarrow 8 + 4 + 2m + n = 11$
 $\therefore \boxed{2m + n = -1} \quad \text{(2)}$

From (1) + (2)

$$\boxed{\begin{matrix} m = 1 \\ n = -3 \end{matrix}}$$

$\therefore P(x) = x^3 + x^2 + x - 3$

now $(x-1)$ is a factor.

$\therefore \boxed{P(x) = (x-1)(x^2 + 2x + 3)}$

$$\begin{array}{r} x^2 + 2x + 3 \\ x-1 \overline{) x^3 + x^2 + x - 3} \\ \underline{x^3 - x^2} \\ 2x^2 + x \\ \underline{2x - 2x} \\ 3x - 3 \end{array}$$

(2.) $f(x) = \sin^{-1}(x-2)$

$\therefore f\left(\frac{3}{2}\right) = \sin^{-1}\left(\frac{3}{2} - 2\right)$
 $= \sin^{-1}\left(-\frac{1}{2}\right)$
 $= \boxed{-\frac{\pi}{6}}$

NB. $|\sin^{-1} \theta| \leq \frac{\pi}{2}$

$$(3) \quad (i) \quad \frac{\binom{24}{5}}{\binom{28}{5}} = \left[\frac{253}{585} \right]$$

$$(ii) \quad \frac{\binom{24}{4} \times \binom{4}{1} + \binom{24}{3} \times \binom{4}{2} + \binom{24}{2} \times \binom{4}{3}}{\binom{28}{5}}$$

$$= \left[\frac{2323}{4095} \right]$$

$$(4) \quad t = \tan x$$

$$\therefore x = \tan^{-1} t$$

$$\left| \frac{dx}{dt} = \frac{1}{1+t^2} \right|$$

$$(5) (i) \quad \int \frac{dx}{\sqrt{25-9x^2}} = \int \frac{dx}{\sqrt{9\left(\frac{25}{9}-x^2\right)}}$$

$$= \frac{1}{3} \int \frac{dx}{\sqrt{\left(\frac{5}{3}\right)^2 - x^2}}$$

$$\left| = \frac{1}{3} \sin^{-1} \frac{3x}{5} + C \right|$$

$$(ii) \quad \int \frac{x+1}{x^2+1} dx = \int \frac{x}{x^2+1} dx + \int \frac{dx}{x^2+1}$$

$$\left| = \frac{1}{2} \ln(x^2+1) + \tan^{-1} x + C \right|$$

$$\begin{aligned}
 (6) \quad \int (e^x + e^{-x})^2 dx &= \int (e^{2x} + 2e^0 + e^{-2x}) dx \\
 &= \int (e^{2x} + 2 + e^{-2x}) dx \\
 &= \frac{1}{2} e^{2x} + 2x - \frac{1}{2} e^{-2x} + C
 \end{aligned}$$

$$\begin{aligned}
 (7). \quad V &= \pi \int_0^2 (e^x)^2 dx \\
 &= \pi \int_0^2 e^{2x} dx \\
 &= \pi \left[\frac{1}{2} e^{2x} \right]_0^2 \\
 &= \frac{\pi}{2} (e^4 - e^0) \\
 &= \frac{\pi}{2} (e^4 - 1) \text{ m}^3
 \end{aligned}$$

Section C (20 marks)

$$1. \int_{\sqrt{2}}^{\sqrt{3}} \frac{dx}{\sqrt{3-4x^2}}$$

$$= \int_{\sqrt{2}}^{\sqrt{3}} \frac{dx}{\sqrt{4(\frac{3}{4}-x^2)}}$$

$$= \frac{1}{2} \int_{\sqrt{2}}^{\sqrt{3}} \frac{dx}{\sqrt{\frac{3}{4}-x^2}}$$

$$= \frac{1}{2} \left[\sin^{-1} \left(\frac{2x}{\sqrt{3}} \right) \right]_{\sqrt{2}}^{\sqrt{3}}$$

$$= \frac{1}{2} (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{4}$$

2. $y = \left(\frac{1}{\ln 2} \right) \ln x$ is curve

$$y' = \frac{1}{\ln 2} \cdot \frac{1}{x}$$

At $P(x_p, y_p)$ $y' = \frac{1}{(\ln 2)x_p}$ is gradient of curve at P .

But gradient of line = $\frac{1}{4 \ln 2}$

$$\frac{1}{(\ln 2)x_p} = \frac{1}{4 \ln 2}$$

$$\Rightarrow x_p = 4 \quad \text{Sub into C} \Rightarrow y_p = \frac{1}{\ln 2} \ln 4$$

$$y_p = 2$$

$$\therefore P = (4, 2)$$

3. Prove $(9^n - 1)$ is divisible by 8, $n \in \mathbb{Z}^+$
let $n=1$, then $9^1 - 1 = 8$ which is divis. by 8

$\therefore$ true for $n=1$

Assume true for $n=k$

i.e. let $9^k - 1 = 8m \quad \exists m \in \mathbb{Z}^+$

Let $n=k+1$ Must show $9^{k+1} - 1 = 8p \quad \exists p \in \mathbb{Z}$

Now $9^{k+1} - 1$

$$= 9^k \cdot 9 - 1$$

$$= 9(9^k - 1) + 8$$

$$= 9(8m) + 8 \quad \text{from assumption}$$

$$= 8(9m + 1)$$

$$= 8p \quad \exists p \in \mathbb{Z}^+$$

$\therefore$ true for $n=k+1$

Therefore, by the Principle of Mathematical Induction, statement is true for all $n, n \in \mathbb{Z}^+$.

3

$$\frac{dy}{dx} = 2e^{2x} - 3e^x \text{ and } y\text{-int} = (0,0)$$

Then $y = e^{2x} - 3e^x + C$

$$(0,0) \Rightarrow 0 = 1 - 3 + C$$

$$\Rightarrow C = 2$$

$$\therefore y = e^{2x} - 3e^x + 2$$

For x-intercepts, let $y = 0$

$$\Rightarrow e^{2x} - 3e^x + 2 = 0 \quad e^x \quad \times \quad e^{-x}$$

$$(e^x - 2)(e^x - 1) = 0$$

$$e^x = 1 \text{ or } e^x = 2$$

$$\underline{x = 0} \text{ or } x = \ln 2$$

$$\begin{aligned} \therefore \text{Area} &= \left| \int_0^{\ln 2} (e^{2x} - 3e^x + 2) dx \right| \\ &= \left| \left[\frac{e^{2x}}{2} - 3e^x + 2x \right]_0^{\ln 2} \right| \\ &= \left| (2 - 6 + 2\ln 2) - \left(\frac{1}{2} - 3 + 0 \right) \right| \\ &= \left| -\frac{3}{2} + \ln 4 \right| \\ &= \frac{3}{2} - \ln 4 \text{ square units} \end{aligned}$$

5. (i) $y = x^2 \ln x$

$$y' = x^2 \frac{1}{x} + (\ln x) 2x$$

$$= x + 2x \ln x$$

(ii) Now $\int (x + 2x \ln x) dx = x^2 \ln x + C$

$$\Rightarrow \frac{x^2}{2} + 2 \int x \ln x dx = x^2 \ln x + C$$

$$\int x \ln x dx = \frac{1}{2} \left(x^2 \ln x - \frac{x^2}{2} \right) + C$$

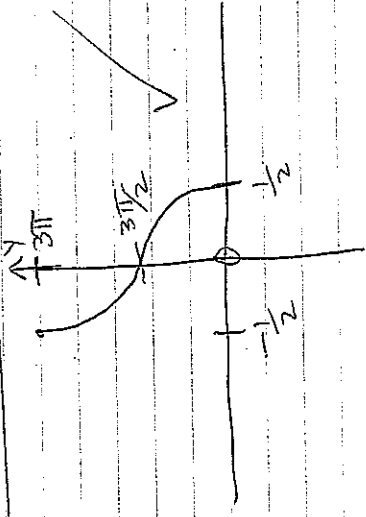
$$\therefore \int (x \ln x) dx = \frac{x^2}{2} \left(\ln x - \frac{1}{2} \right) + C$$

6. $y = 3 \cos^{-1} 2x$

$$\Rightarrow \frac{y}{3} = \cos^{-1} 2x$$

D: $-1 \leq 2x \leq 1$ and Range: $0 \leq \frac{y}{3} \leq \pi$

O: $-\frac{1}{2} \leq x \leq \frac{1}{2}$ $0 \leq y \leq 3\pi$



$$\begin{aligned}
 \text{(i)} \quad x^2 &= 4ay \\
 \therefore 2x &= 4a y' \\
 \therefore y' &= \frac{x}{2a}
 \end{aligned}$$

$$\begin{aligned}
 \text{At } P, y' &= \frac{2ap}{2a} \\
 &= p
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Eqn of tangent is} \\
 y - ap^2 &= p(x - 2ap) \\
 y - ap^2 &= px - 2ap^2 \\
 y &= px - ap^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Eqn of tangent at } Q \text{ is} \\
 y &= qx - aq^2 \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad px - ap^2 &= qx - aq^2 \\
 \therefore (p - q)x &= ap^2 - aq^2 \\
 \therefore (p - q)x &= a(p - q)(p + q) \\
 \therefore x &= a(p + q)
 \end{aligned}$$

$$\begin{aligned}
 \therefore y &= p(a(p + q)) - ap^2 \\
 &= apq
 \end{aligned}$$

$$\therefore T \text{ is } (a(p + q), apq) \quad (2)$$

$$\begin{aligned}
 \text{(iii)} \quad M \text{ is } \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) \\
 = \left(a(p + q), \frac{a(p^2 + q^2)}{2} \right) \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad R \text{ is } \left(\frac{a(p + q) + a(p + q)}{2}, \right. \\
 \left. \frac{apq + \frac{a(p^2 + q^2)}{2}}{2} \right)
 \end{aligned}$$

$$= \left(a(p + q), \frac{a(2pq + p^2 + q^2)}{4} \right)$$

$$= \left(a(p + q), \frac{a(p + q)^2}{4} \right) \text{ as given} \quad (1)$$

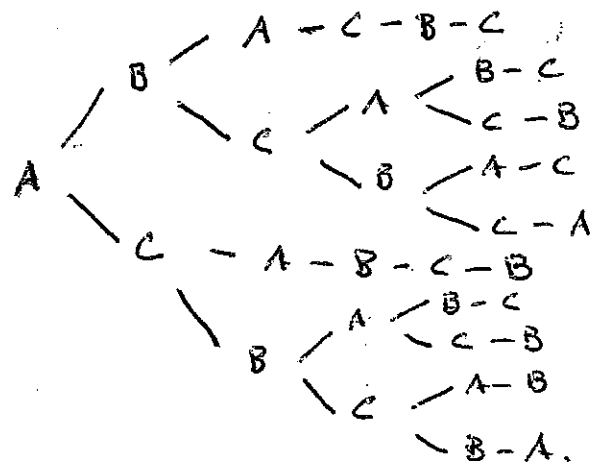
$$\text{(v)} \quad \text{Let } x = a(p + q)$$

$$\therefore y = \frac{x^2}{a} \cdot \frac{1}{4}$$

$$x^2 = 4ay \quad (2)$$

$\therefore R$ lies on the parabola $x^2 = 4ay$

2



10 arrangements.

$\therefore$ Total arrangements

$$= 10 \times 3$$

$$= 30 \quad (3)$$

$$3 \text{ (i) } (a+b)(a^{n-1}+b^{n-1}) - ab(a^{n-2}+b^{n-2})$$

$$= a^n + ab^{n-1} + ba^{n-1} + b^n - a^{n-1}b - ab^{n-1}$$

$$= a^n + b^n$$

$$(ii) x^{k+2} + \frac{1}{x^{k+2}} = \left(x + \frac{1}{x}\right) \left(x^{k+1} + \frac{1}{x^{k+1}}\right)$$

$$- 1 \cdot \left(x^k + \frac{1}{x^k}\right)$$

$$= 2 \cos \theta \cdot 2 \cos(k+1)\theta - 2 \cos k\theta$$

$$= 2 \cos \theta (2 \cos k\theta \cos \theta - 2 \sin k\theta \sin \theta)$$

$$- 2 \cos k\theta$$

$$= 4 \cos k\theta \cos^2 \theta - 4 \sin k\theta \sin \theta \cos \theta$$

$$- 2 \cos k\theta$$

$$= 2 \cos k\theta (2 \cos^2 \theta - 1) - 2 \sin k\theta \sin 2\theta$$

$$= 2 (\cos k\theta \cos 2\theta - \sin k\theta \sin 2\theta)$$

$$= 2 \cos(k+2)\theta \quad (3)$$

$$(iii) S(n) = (x + x^3 + \dots + x^{2n-1}) + \left(\frac{1}{x} + \frac{1}{x^3} + \dots + \frac{1}{x^{2n-1}}\right)$$

$$= \frac{\sin 2n\theta}{\sin \theta}$$

Step 1: shows $S(1)$ is true

$$\text{i.e. } x + \frac{1}{x} = \frac{\sin 2\theta}{\sin \theta}$$

$$\text{LHS} = 2 \cos \theta$$

$$\text{RHS} = \frac{2 \sin \theta \cos \theta}{\sin \theta}$$

$$= 2 \cos \theta$$

$\therefore S(1)$ is true

Step 2: Assume $S(k)$ is true

$$\text{i.e. } (x + x^3 + \dots + x^{2k-1}) + \left(\frac{1}{x} + \frac{1}{x^3} + \dots + \frac{1}{x^{2k-1}}\right)$$

$$= \frac{\sin 2k\theta}{\sin \theta}$$

show $S(k+1)$ is true

$$\text{i.e. } (x + x^3 + \dots + x^{2k+1}) + \left(\frac{1}{x} + \frac{1}{x^3} + \dots + \frac{1}{x^{2k+1}}\right)$$

$$= \frac{\sin(2k+1)\theta}{\sin \theta}$$

$$\text{LHS} = \frac{\sin 2k\theta}{\sin \theta} + x^{2k+1} + \frac{1}{x^{2k+1}}$$

$$= \frac{\sin 2k\theta}{\sin \theta} + 2 \cos(2k+1)\theta$$

$$= \frac{\sin 2k\theta + 2 \cos(2k+1)\theta \sin \theta}{\sin \theta}$$

$$= \frac{\sin 2k\theta + 2 \sin \theta (\cos k\theta \cos \theta - \sin k\theta \sin \theta)}{\sin \theta}$$

$$= \frac{\sin 2k\theta + \cos k\theta \cdot 2 \sin \theta \cos \theta - 2 \sin^2 \theta \sin k\theta}{\sin \theta}$$

$$= \frac{\cos 2k\theta \cdot \sin 2\theta + \sin 2k\theta (1 - 2 \sin^2 \theta)}{\sin \theta}$$

$$= \frac{\cos 2k\theta \sin 2\theta + \sin 2k\theta \cos 2\theta}{\sin \theta}$$

$$= \frac{\sin(2k+2)\theta}{\sin \theta}$$

$$= \text{RHS}$$

$\therefore$ if $S(k)$ is true, $S(k+1)$ is true

Step 3: $S(1)$ is true and $S(k+1)$ is true if $S(k)$ is true.

$\therefore$ By the process of Mathematical

Induction $S(n)$ is true for all integral $n \geq 1$. (4)